

5

Newborn babies can be fed with breast milk or with formula milk. Both types of milk contain carbohydrates, lipids and proteins.

- Human breast milk also contains a bile-activated lipase. This enzyme is thought to be inactive in milk but activated by bile in the small intestine of the newborn baby.
- Formula milk does not contain a bile-activated lipase.

Scientists investigated the benefits of breast milk compared with formula milk.

0 5 . 1

The scientists used kittens (newborn cats) as model organisms in their laboratory investigation.

Other than ethical reasons, suggest **two** reasons why they chose to use cats as model organisms.

[2 marks]

1 _____

2 _____

0 5 . 2

Before starting their experiments, the scientists confirmed that, like human breast milk, cat's milk also contained bile-activated lipase.

To do this, they added bile to cat's milk and monitored the pH of the mixture.

Explain why monitoring the pH of the mixture could show whether the cat's milk contained lipase.

[2 marks]

Question 5 continues on the next page

The scientists then took 18 kittens. Each kitten had been breastfed by its mother for the previous 48 hours.

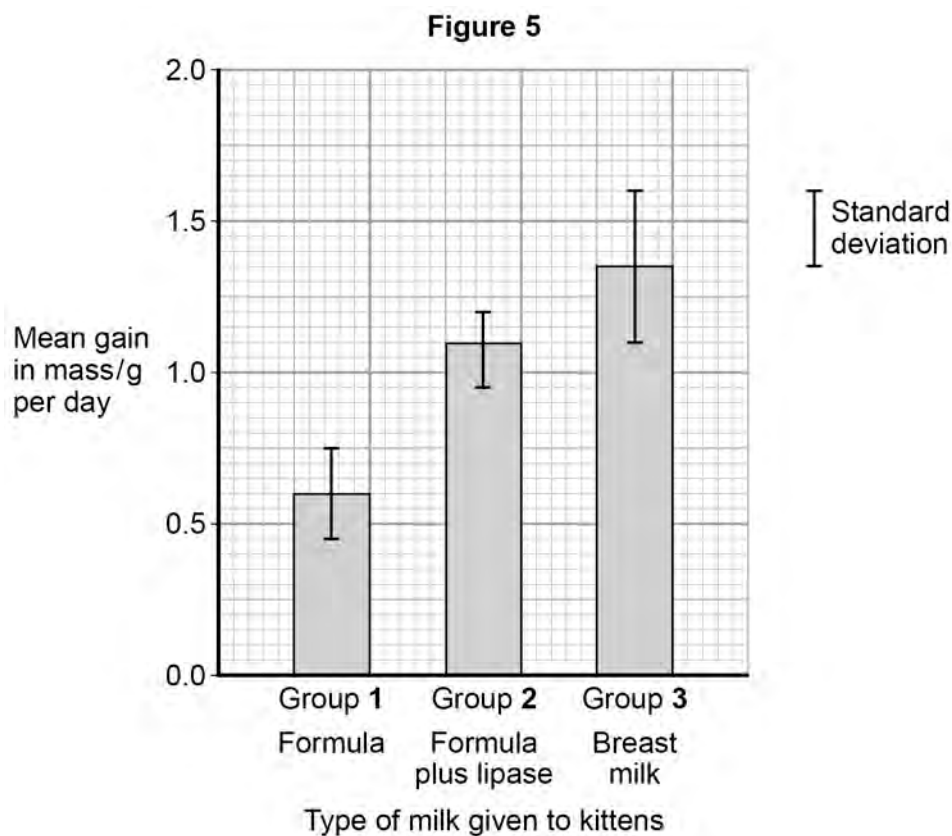
The scientists divided the kittens randomly into three groups of six.

- The kittens in group 1 were fed formula milk.
- The kittens in group 2 were fed formula milk plus a supplement containing bile-activated lipase.
- The kittens in group 3 were fed breast milk taken from their mothers.

Each kitten was fed 2 cm³ of milk each hour for 5 days.

The scientists weighed the kittens at the start of the investigation and on each day for 5 days.

Figure 5 shows the scientists' results.



0 5 . 3 What can you conclude from **Figure 5** about the importance of bile-activated lipase in breast milk?

[3 marks]

[Extra space]

Turn over for the next question

Turn over ►